

Arduino-Based Environmental Monitor

Project Presentation & Technical Overview



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Arduino-Based Environmental Monitor

Real-time multisensor telemetry for automated environmental awareness & hazard detection



Temp & Humidity

DHT11 Sensor

Tracks ambient climate conditions with high precision.

- **Purpose:** Climate stability & HVAC automation
- **Metrics:** °C / °F and Relative Humidity (%RH)
- **Output:** Digital signal to Arduino MCU



Rain Detector

FC-37 Rain Board

Detects precipitation presence and intensity via nickel-coated tracks.

- **Purpose:** Weather sensing & roof automation
- **Metrics:** Resistance drop upon water contact
- **Output:** Digital trigger & Analog depth level



Smoke & Gas

MQ-2 Gas Sensor

Monitors air quality and detects flammable gas or smoke threats.

- **Purpose:** Early fire warning & safety alerts
- **Metrics:** LPG, Smoke, CO, Alcohol concentration
- **Output:** Threshold alert & Analog PPM value

How the System Works

End-to-end data flow from environmental sensing to automated action

01

1. Data Sensing

Environmental Inputs

Sensors continuously monitor physical environmental attributes.

- **DHT11:** Climate metrics
- **FC-37:** Water contact
- **MQ-2:** Gas concentration

02

2. Processing

Arduino Microcontroller

MCU receives raw signals, converts ADC values, and evaluates logic.

- **Signal Sampling:** 1 Hz
- **Threshold Check:** Limits
- **Telemetry:** Data prep

03

3. Transmission

IoT Gateway / Wi-Fi

Telemetry payload is dispatched wirelessly to local or cloud hubs.

- **Protocol:** MQTT / HTTP
- **Payload:** JSON metrics
- **Latency:** Real-time

04

4. Action & Alert

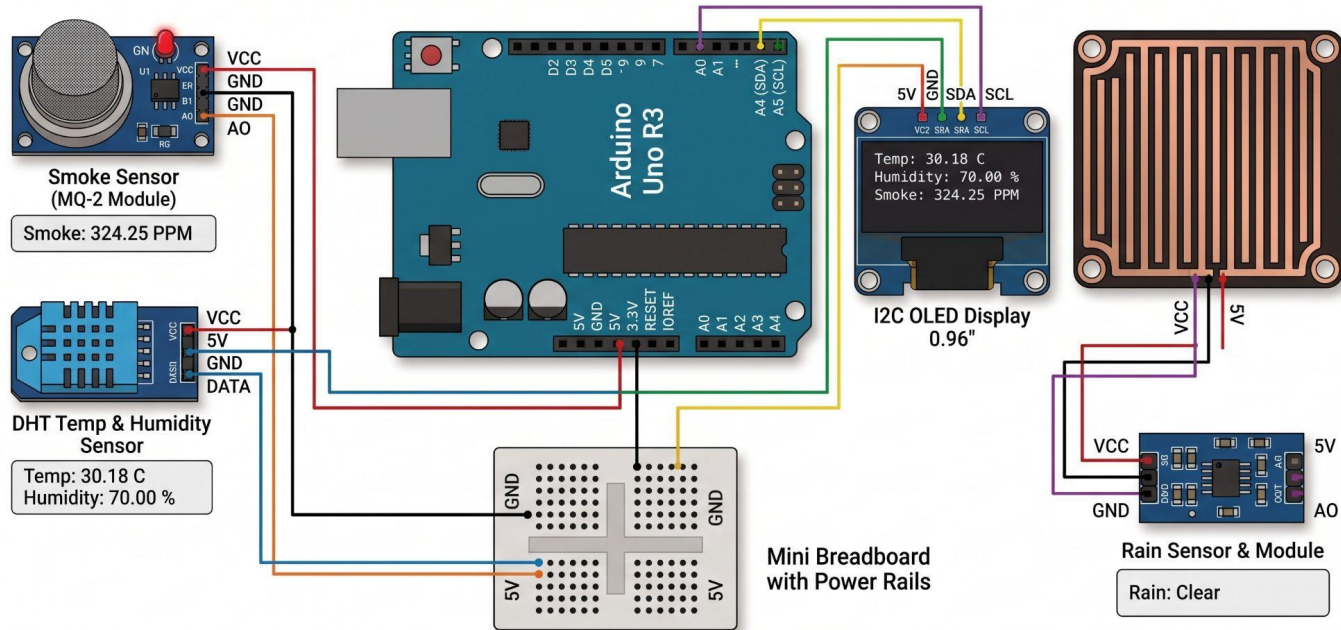
Automation & UI

Dashboards update and automated safety hardware triggers instantly.

- **Alerts:** Push & Buzzers
- **Control:** Relays / Fans
- **Dashboard:** Live UI

Hardware Circuit Diagram

Environmental Monitoring Station Circuit Diagram



Questions & Discussion

Feel free to ask any questions about the Arduino Environmental Monitor project



Technical Specs

Hardware & Sensors

Inquiries regarding MCU processing, sensor calibration, or circuit layout.

- DHT11, MQ-2 & FC-37 integration
- Power efficiency & pinout



System Logic

Data & Firmware

Questions on sampling rates, signal filtering, and alert thresholds.

- ADC conversion algorithms
- Real-time telemetry pipeline



Applications

Use Cases & Scale

Discussion on industrial deployment, automation, and future expansions.

- HVAC & hazard safety
- IoT cloud dashboarding

CONCLUSION

Thank You!

Arduino-Based Environmental Monitoring System Presentation